FlowTrack Transformation Brief

Project Evolution: From Internal IMS to Multi-Tenant SaaS Platform

📋 Executive Summary

Project: FlowTrack (formerly IMS-AP - Inventory Management System)
Date: February 2026
Phase: Phase 2 - SaaS Platform Transformation
Branch: dev-Phase-2

The Pivot:

FlowTrack is transitioning from a single-organization internal inventory management system to a **multi-tenant SaaS platform** that connects retailers, vendors, and investors in a supply chain and investment ecosystem.

🔄 Role Structure Transformation

OLD MODEL (Academic Project - Single Organization)

Role	Purpose	Problem
`ROLE_ADMIN`	System administrator for one organization	Assumes single-tenant usage
`ROLE_MANAGER`	Inventory manager with elevated privileges	Too granular for small businesses
`ROLE_STAFF`	Staff member with limited access	Unnecessary complexity for SMBs
`ROLE_USER`	Basic user with view-only access	Vague purpose, minimal functionality

Assumptions of Old Model:

- FlowTrack is deployed **within** one organization (e.g., a university)
- Multiple employees have different access levels
- Designed for hierarchical organizational structure
- Items are lent out to borrowers (library-style system)

Why This Failed:

- ❌ Not scalable for multiple independent businesses
- ❌ Small-mid businesses don't need complex employee hierarchies
- ❌ No multi-tenancy support
- ❌ No business-to-business interactions
- ❌ No investment/financial ecosystem

NEW MODEL (SaaS Platform - Multi-Tenant)

Role	Purpose	Use Case	Account Type
`ROLE_PLATFORM_ADMIN`	FlowTrack team managing the entire platform	System management, user verification, dispute resolution	Internal (FlowTrack staff)
`ROLE_RETAILER`	Small-mid scale business owners	Inventory management, supplier connections, investment requests	Customer
`ROLE_VENDOR`	Product suppliers and distributors	Supply products to retailers, manage orders	Customer
`ROLE_INVESTOR`	Individual or institutional investors	Provide capital to retailers, track ROI	Customer

Key Architectural Changes:

1. ☑ **Multi-tenancy:** Each retailer = separate account with isolated data
2. ☑ **B2B Transactions:** Retailers ↔ Vendors (product supply)
3. ☑ **Investment Ecosystem:** Retailers ↔ Investors (capital funding)
4. ☑ **Identity Verification:** All users must prove their identity before access
5. ☑ **Open Registration:** Users self-register, platform admin approves

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👥 Detailed Role Analysis

1. ROLE_PLATFORM_ADMIN

Who are they?

- The FlowTrack development/operations team
- System administrators of the SaaS platform

Responsibilities:

- Approve/reject new user registrations
- Verify identity documents submitted by retailers/vendors/investors
- Manage platform-wide settings
- Resolve disputes between parties
- Monitor system health and security
- Handle payment gateway issues (Razorpay integration)
- Access to all data for audit purposes

Why we need this:

- Someone must manage the platform itself
- Identity verification requires manual review
- Trust & safety for the ecosystem

Dashboard Features:

- Pending verification queue (with uploaded documents)
- User management (approve, suspend, ban)
- Platform analytics (total users, transactions, revenue)
- System configuration

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2. ROLE_RETAILER

Who are they?

- Small to mid-scale business owners (5-50 employees)
- Shop owners, e-commerce businesses, restaurants, pharmacies, etc.
- Target: Businesses needing inventory tracking & investment

Business Problem They Face:

1. **Inventory Management:** Tracking stock levels, reorders, product flow
2. **Supplier Relationships:** Finding reliable vendors, managing orders
3. **Capital Shortage:** Need investment for inventory expansion
4. **Cash Flow:** Tied up capital in inventory, need working capital loans

What FlowTrack Solves:

- ☑ Real-time inventory tracking
- ☑ Direct connection to verified suppliers (vendors)
- ☑ Access to investors willing to fund inventory
- ☑ Transaction management (orders, payments via Razorpay)
- ☑ Analytics (sales forecasting, stock optimization)

Why Identity Verification Matters:

- Investors need to know the business is legitimate
- Vendors need trust before extending credit
- Prevents fraud and fake accounts
- Legal compliance for financial transactions

Registration Information Needed:

- **Business Identity:**
  - Business name (as per registration)
  - Business type (Retail Store, E-commerce, Restaurant, Pharmacy, etc.)
  - Business registration number (mandatory in India)
  - GST number (for businesses > ₹20 lakh turnover)
  - Trademark (optional, for brand identity)
- **Owner Identity:**
  - Owner's full name (first + last)
  - Email (for account access)
  - Phone number (for OTP verification)

    - Username & password

- **Business Address:**
    - Full business address (for verification)
- **Documents (Proof):**
    - **Business License/Shop Act License** (proves business exists)
    - **Owner's ID Proof** (Aadhar/PAN/Passport)
    - Optional: GST Certificate, Bank Statement (last 3 months)

Why Each Piece Matters:

- **Business Registration Number:** Verifiable with government databases
- **GST Number:** Cross-check with GSTN portal, proves tax compliance
- **Address:** Physical verification possible if needed
- **ID Proof:** Links business to real person (accountability)
- **Bank Statement:** Proves financial activity, credibility for investors

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3. ROLE_VENDOR

Who are they?

- Manufacturers, wholesalers, distributors
- Product suppliers (stationary, electronics, furniture, etc.)
- B2B sellers looking for retailer customers

Business Problem They Face:

1. **Customer Acquisition:** Hard to find new retailers to supply
2. **Payment Delays:** Retailers don't pay on time
3. **Inventory Uncertainty:** Don't know what retailers need
4. **Trust Issues:** Fear of non-payment from unknown retailers

What FlowTrack Solves:

- ☑ Marketplace to find verified retailers
- ☑ Secure payment through Razorpay (escrow-like)
- ☑ Order management system
- ☑ Demand visibility (what retailers are searching for)
- ☑ Credit scoring (retailers with good payment history)

Why Identity Verification Matters:

- Retailers need verified, legitimate suppliers
- Payment disputes require legal entity identification
- Prevents counterfeit/illegal product supply
- Tax compliance (GST for B2B transactions)

Registration Information Needed:

- **Company Identity:**
  - Company name (official registered name)
  - Business type (Manufacturer, Wholesaler, Distributor, Agent)
  - Trade License Number
  - GST Number (mandatory for B2B in India)
  - PAN Number (for tax filing)
- **Contact Person:**
  - Authorized person's name
  - Email & phone
  - Username & password
- **Product Information:**
  - Product categories offered (Electronics, Furniture, Chemicals, etc.)
  - Company description
- **Banking Information (for payments):**
  - Bank account number
  - Bank name
  - IFSC code
  - Account holder name (must match company name)
- **Documents (Proof):**
  - **Trade License** (proves legal business operation)
  - **GST Certificate** (mandatory for B2B sales)
  - **Company Registration Certificate** (ROC document)
  - Optional: Product samples, certifications (ISO, etc.)

Why Each Piece Matters:

- **Trade License:** Legal authority to do business
- **GST Certificate:** Cross-verify with GSTN, enables invoicing
- **Bank Details:** Direct payment, no cash = transparency
- **Company Registration:** Proves legal entity, can be sued if fraud
- **Product Categories:** Helps retailers find right suppliers

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4. ROLE_INVESTOR

Who are they?

- Individual investors (angel investors, wealthy individuals)
- Institutional investors (venture capital, private equity firms)
- Family offices, investment clubs

Business Problem They Face:

1. **Deal Flow:** Hard to find good investment opportunities
2. **Due Diligence:** Expensive to verify businesses
3. **Monitoring:** Can't track how invested money is used
4. **Risk Management:** Fear of fraud or business failure

What FlowTrack Solves:

- ☑ Pre-verified retailer businesses (we do KYC)
- ☑ Transparent financials (inventory value, sales data)
- ☑ Investment tracking (see how money is deployed)
- ☑ ROI monitoring (real-time business performance)
- ☑ Diversification (invest in multiple retailers)

Why Identity Verification Matters:

- Regulatory compliance (SEBI regulations for investments)
- Anti-money laundering (AML) requirements
- Tax compliance (interest/capital gains reporting)
- Legal recourse if disputes arise

Registration Information Needed:

- **Investor Identity:**
  - Investor name (individual name or company name)
  - Investor type (Individual, Angel Investor, VC, PE, Family Office)
  - PAN Number (mandatory for financial transactions in India)
  - Aadhar Number (for individual investors)
  - Company registration number (for institutional investors)
- **Contact Information:**
  - Email & phone
  - Registered address
  - Username & password
- **Investment Profile:**
  - Investment capacity (available capital)
  - Minimum investment amount (₹50,000? ₹5 lakh?)
  - Maximum investment amount (per deal)
  - Preferred sectors (Retail, F&B, E-commerce, Healthcare, etc.)
  - Investment experience (years)
  - Risk appetite (Low/Medium/High)
  - Expected ROI percentage (10%? 20%? 30%?)
- **Banking Information:**
  - Bank account number (for disbursements)
  - Bank name
  - IFSC code

- **Documents (Proof):**
  - **PAN Card** (mandatory, verifiable with IT department)
  - **Aadhar Card** (for individuals)
  - **Bank Statement** (last 6 months - proves financial capacity)
  - **Investment Portfolio** (optional - shows track record)
  - **Net Worth Certificate** (for large investors)
  - **Company Registration** (for institutional investors)

Why Each Piece Matters:

- **PAN Number:** Tax compliance, verifiable, unique identity
- **Bank Statement:** Proves they have money to invest (no fake investors)
- **Investment Profile:** Match investors with suitable opportunities
- **Risk Appetite:** Don't show high-risk retailers to conservative investors
- **Expected ROI:** Align expectations, prevent disputes

🎯 Target Market Analysis

Small-Mid Scale Businesses (Retailers)

Size:

- 1-50 employees
- Annual revenue: ₹10 lakh - ₹10 crore
- Single location or 2-5 branches

Pain Points:

1. Manual inventory tracking (Excel sheets)
2. Cash flow problems (money tied in inventory)
3. Difficulty getting bank loans (lack of collateral)
4. Supplier payment terms (30-60 days credit)
5. No visibility into demand forecasting

Why They Need FlowTrack:

- Professional inventory system (affordable vs custom software)
- Access to working capital (from investors)
- Verified supplier network (better prices, trust)
- Growth enabler (scale with investment, not just loans)

🔐 Identity Verification Strategy

Why Identity Verification is Critical:

1. **Financial Transactions:**
   - Razorpay integration requires KYC for merchants

    - Investment = regulated activity (SEBI compliance)
    - Tax reporting (TDS, GST, income tax)

2. **Trust & Safety:**
    - Prevents fake businesses
    - Reduces fraud (payment defaults, scams)
    - Legal accountability (can pursue if issues)

3. **Business Credibility:**
    - Verified badge increases trust
    - Better match-making (retailers ↔ vendors)
    - Higher investment success rate

Verification Workflow:

```
User Registers → Uploads Documents → Status: PENDING
         ↓
Platform Admin Reviews → Verifies Documents → Cross-checks with Gov Databases
         ↓
Decision: APPROVED or REJECTED (with reason)
         ↓
Email Notification → User can login (if approved)
```

Government Database Cross-Verification (India):

- **GST Number:** GSTN portal (free API)
- **PAN Number:** Income Tax e-filing portal
- **Company Registration:** MCA (Ministry of Corporate Affairs)
- **Trade License:** State government portals
- **Aadhar:** UIDAI (requires consent, paid API)

📊 Information Collection Impact Analysis

For Retailers:

Information	Verification Method	Impact on Platform	Risk if Not Collected
Business Registration Number	MCA/State Portal	Proves legal business	Fake businesses, fraud
GST Number	GSTN API	Tax compliance, invoicing	Illegal operations
Business License	Manual review	Legal to operate	Unlicensed entities
Owner ID Proof	Aadhar/PAN	Links to real person	Anonymous accounts

Information	Verification Method	Impact on Platform	Risk if Not Collected
Bank Details	Razorpay KYC	Payment processing	Can't disburse funds
Business Address	Google Maps verify	Physical presence	Virtual scams

For Vendors:

Information	Verification Method	Impact on Platform	Risk if Not Collected
GST Certificate	GSTN API	B2B invoicing	Tax evasion
Trade License	State commerce dept	Legal supplier	Counterfeit products
Bank Account	Penny-drop verification	Payment routing	Payment fraud
Company Registration	MCA database	Legal entity	Fly-by-night operators
Product Categories	Manual verification	Better matching	Misrepresentation

For Investors:

Information	Verification Method	Impact on Platform	Risk if Not Collected
PAN Number	IT portal	Tax compliance	Income tax issues
Bank Statement	Analyze for capacity	Proves solvency	Fake investors
Net Worth Certificate	CA verification	Investment limits	Over-commitment
Aadhar/Passport	UIDAI/Passport office	Identity proof	Identity theft
Investment Experience	Self-declared + verify	Risk assessment	Unsuitable investments

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🚀 Technical Implementation

Current Status (as of Feb 25, 2026):

☑ Completed:

- Role entity refactored (4 new roles)
- SecurityConfig updated with role-based routing
- DataInitializer creates new roles on startup
- Landing page with role selection UI
- Profile entities created (RetailerProfile, VendorProfile, InvestorProfile)
- Registration DTOs with validation
- Registration controllers with routing
- Database schema designed

⏳ Pending:

- Registration HTML forms (3 pages)

- Profile repositories
- Registration service layer
- Document upload functionality (file storage)
- Email notification service (SMTP)
- Verification dashboard for Platform Admin
- Role-specific dashboards (4 pages)
- Razorpay integration
- Government API integrations for verification

🔍 Research Questions for Registration Requirements

For GPT/AI to Research:

1. **Legal Compliance (India):**
    - What documents are **legally required** for B2B marketplace registration?
    - SEBI regulations for investment platforms (if applicable)?
    - RBI guidelines for payment aggregation?
    - GST requirements for platform service fee collection?
2. **Security Best Practices:**
    - What information should we **hash/encrypt** (PAN, Aadhar)?
    - Document storage: S3 with encryption? Retention period?
    - Access logs for sensitive data viewing?
    - GDPR-like compliance (even though India doesn't have full GDPR)?
3. **UX vs Security Trade-off:**
    - How much information is **too much** for registration?
    - Can we make some fields optional initially (progressive profiling)?
    - Verification at registration vs post-registration?
    - Mobile number OTP vs email verification?
4. **Financial Information:**
    - Should we ask for **income/revenue** from retailers?
    - Bank statement: last 3 months or 6 months?
    - Net worth certificate: mandatory or optional for investors?
    - Credit score: integrate with CIBIL/Experian?
5. **Competitive Analysis:**
    - What does **IndiaMART** ask during vendor registration?
    - What does **Zerodha** ask for investor KYC?
    - What does **Amazon Business** ask from retailers?
    - What does **ShopX/Udaan** ask in their B2B platform?
6. **Fraud Prevention:**

    - Should we require **video KYC** (RBI compliance)?
    - IP address logging + geolocation matching?
    - Device fingerprinting for account security?
    - Facial recognition for ID proof matching?

7. **Operational Feasibility:**
    - How long should **manual verification** take? (24hrs, 48hrs, 7 days?)
    - Should we have auto-approval for certain criteria?
    - What if documents are in regional languages (Hindi, Tamil, etc.)?
    - Court-issued documents vs self-attested copies?

💡 Recommendations for Next Steps

Immediate (This Week):

1. ☑ Research government API availability (GST, PAN verification)
2. ☑ Finalize mandatory vs optional fields for each role
3. ☑ Design 3 registration forms (HTML/Thymeleaf)
4. ☑ Implement file upload (Spring MultipartFile)

Short-term (Next 2 Weeks):

1. ☑ Build verification dashboard for Platform Admin
2. ☑ Integrate email notifications (Spring Mail + Gmail SMTP)
3. ☑ Create approval/rejection workflow
4. ☑ Build role-specific dashboards

Medium-term (1 Month):

1. ☑ Razorpay integration (test mode first)
2. ☑ Retailer ↔ Vendor transaction flow
3. ☑ Investment request system
4. ☑ Reporting & analytics

Long-term (2-3 Months):

1. ☑ Government API integrations (automated verification)
2. ☑ Mobile app (optional)
3. ☑ AI-powered matching (retailers ↔ vendors, retailers ↔ investors)
4. ☑ Deploy on AWS with domain

📊 Success Metrics

For Platform:

- Number of verified users (by role)
- Retailer-Vendor transactions (volume & value)

- Investment deals closed
- Platform service fee revenue

For Retailers:

- Inventory turnover improvement
- Capital raised via platform
- Supplier network expansion

For Vendors:

- Number of retailer connections
- Order volume growth
- Payment cycle reduction

For Investors:

- ROI achieved
- Number of deals
- Portfolio diversification

🎓 Learning Outcomes (for GPT Context)

This project demonstrates:

1. **Multi-tenant SaaS architecture** (isolation, scalability)
2. **Identity verification systems** (KYC/KYB)
3. **Role-based access control** (Spring Security)
4. **Payment gateway integration** (Razorpay)
5. **B2B marketplace mechanics** (supply chain)
6. **Investment platform basics** (fintech)
7. **Government API integration** (verification)
8. **Email notification systems** (transactional emails)
9. **Document management** (upload, storage, security)
10. **Real-world compliance** (GST, tax, regulations)

❓ Key Questions to Answer via Research

When researching registration requirements, focus on:

1. **What is the MINIMUM viable information** needed for each role to:
    - Establish trust between parties?
    - Enable transactions?
    - Comply with Indian law?
2. **What is the MAXIMUM acceptable friction** during registration before users abandon?
3. **Which verifications can be automated** vs manual review?

4. **What are the legal liabilities** if we don't collect certain information (e.g., money laundering, tax evasion)?

5. **How do similar Indian platforms** handle this?

    - B2B: Udaan, IndiaMART, Moglix
    - Investment: Angellist India, LetsVenture
    - Payment: Razorpay, Paytm Business

End of Brief

This document serves as context for AI-assisted research on registration requirements and identity verification standards for a B2B marketplace + investment platform in India.